

The spectrum of limit models in stable AECs

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Based largely on joint work with Marcos Mazari-Armida

1. Background - AECs
2. Background - Limit models
3. Papers 1 ([BeMa]) and 2 ([Bea25b]) - The spectrum of limit models
4. Paper 3 ([Bea25a]) - Extending to 'weak' independence relations
5. Paper 4 ([Bea26]) - Disjoint non-forking amalgamation

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The cost of the more general setting is that we lose useful tools - in particular, formulas and compactness.

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4. (Tarski-Vaught chain axiom) For all $\leq_{\mathbf{K}}$ -increasing chains $\langle M_i : i < \delta \rangle$
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Notation

$\mathbf{K}_{\lambda}$ is the subclass of $\mathbf{K}$ of models of size λ .

Background - Examples of AECs

Examples

1. $\mathbf{K} = (\text{Mod}(T), \preceq)$ for a first order theory T , with $\text{LS}(\mathbf{K}) = |\text{L}(T)| + \aleph_0$
2. $\mathbf{K} = (\{M \models T : M \text{ is } \aleph_0\text{-homogeneous and } D(M) = \Gamma\}, \preceq)$ for a first order theory T where $\Gamma \subseteq D(M)$, with $\text{LS}(\mathbf{K}) = |\text{L}(T)| + \aleph_0$
3. $\mathbf{K} = (\text{Mod}(\varphi), \subseteq)$ for a universal sentence $\varphi \in \text{L}_{\infty, \omega}$, with $\text{LS}(\mathbf{K}) = |\text{L}(\varphi)| + \aleph_0$
4. $\mathbf{K} = (\text{Mod}(\varphi), \preceq_F)$ for some sentence $\varphi \in \text{L}_{\infty, \omega}$ and F a fragment containing φ , with $\text{LS}(\mathbf{K}) = |F| + |\text{L}(\varphi)| + \aleph_0$
5. $\mathbf{K} = (\text{Mod}(\varphi), \preceq_F^*)$ for some sentence $\varphi \in \text{L}_{\omega_1, \omega}(Q)$ and F the smallest countable fragment containing φ , with $\text{LS}(\mathbf{K}) = \aleph_1$.
6. To name a few of the many algebraic examples,
 - 6.1 Torsion Abelian groups with $\subseteq$, pure subgroup, or $\preceq$
 - 6.2 Modules over a fixed ring with $\subseteq$, pure submodule, or $\preceq$
 - 6.3 Actions of a fixed monoid with $\subseteq$ or $\preceq$

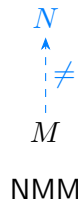
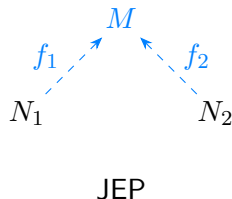
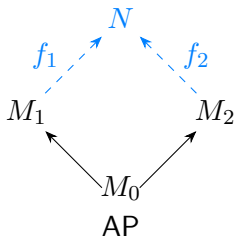
Background - Some notions which lift

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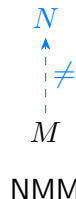
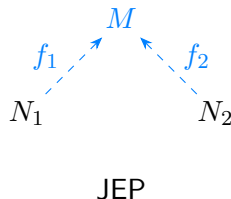
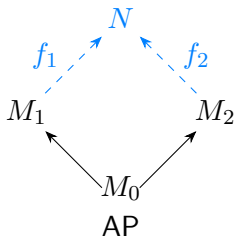
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$(\text{Mod}(T), \preceq)$ satisfies these hypotheses if T is complete with infinite models (but again, our context is much broader). AP in particular makes Galois types behave well.

Background - Extra structural assumptions: Independence relations

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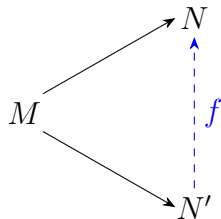
We will normally ask that our independence relations satisfy some collections of the properties of forking in first order stable theories (uniqueness, extension, symmetry, local character, continuity, etc).

Background - Universal extensions

Definition

Let $\mathbf{K}$ be an AEC with $\lambda \geq \text{LS}(\mathbf{K})$.

For $M, N \in \mathbf{K}_\lambda$, N is *universal* over M if $M \leq_{\mathbf{K}} N$ and whenever $N' \in \mathbf{K}_\lambda$ and $M \leq_{\mathbf{K}} N'$, there exists a $\mathbf{K}$ -embedding $f : N' \rightarrow N$ fixing M . We write $M \leq_{\mathbf{K}}^u N$.



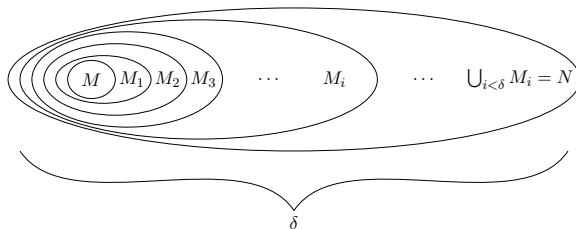
Intuitively, this says N contains a copy of every superstructure N' of M of size λ .

Background - Limit models

Definition

Let $\mathbf{K}$ be an AEC where $\lambda \geq \text{LS}(\mathbf{K})$ and $\delta < \lambda^+$ is a limit ordinal.

$N \in \mathbf{K}$ is a (λ, δ) -limit model over M if there exists a $\leq_{\mathbf{K}}^u$ -increasing sequence $\langle M_i : i < \delta \rangle$ in $\mathbf{K}_\lambda$ where $M_0 = M$ and $N = \bigcup_{i < \delta} M_i$.



Fact (Existence of limit models)

If $\mathbf{K}$ is λ -stable and $\mathbf{K}_\lambda$ has AP, then for any $M \in \mathbf{K}_\lambda$, there exists $N \in \mathbf{K}_\lambda$ universal over M . And for any $\delta < \lambda^+$ limit, there is a (λ, δ) -limit N over M .

Background - Limit models

Fact (Uniqueness up to cofinality)

Suppose $\mathbf{K}$ is a λ -stable AEC and $\mathbf{K}_\lambda$ has AP, $\delta_1, \delta_2 < \lambda^+$ are limits with $\text{cf}(\delta_1) = \text{cf}(\delta_2)$, M is a (λ, δ_1) -limit model over M_0 , and N is a (λ, δ_2) -limit over M_0 . Then $M \cong_{M_0} N$.

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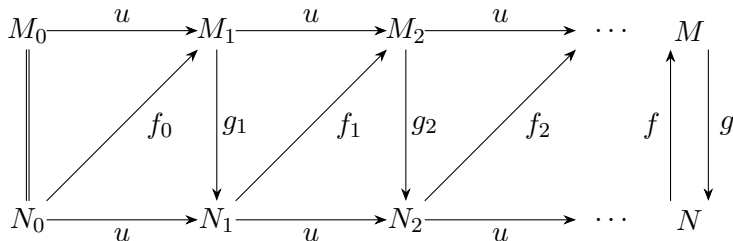
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Superstable - History of limit models

Proving that all limit models (over M) are isomorphic (over M) in certain settings is an important step towards Shelah's categoricity conjecture. Shelah and Villaveces drew up the blueprint of a proof under strongish assumptions [SV99], but there were errors in the proof.

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What about (possibly strictly) stable AECs?

Beyond superstable - Existing result

An improved result result in the strictly stable setting:

Theorem (Boney, VanDieren)

Let $\mathbf{K}$ be an AEC stable in $\lambda \geq \text{LS}(\mathbf{K})$ with AP, JEP, and NMM in $\mathbf{K}_\lambda$. Suppose $\kappa \leq \theta < \lambda^+$ are regular, and that the independence relation λ -non-splitting, satisfies $(\geq \kappa)$ -local character, universal continuity, and (λ, θ) -symmetry. If $M, N_1, N_2 \in K_\lambda$ where N_l is a (λ, δ_l) -limit over M and $\text{cf}(\delta_l) \geq \kappa$ for $l = 1, 2$, then there is an isomorphism from N_1 to N_2 fixing M .

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So we should expect 'long' limits to be isomorphic. But what happens for a general independence notion, and what happens for limit models 'shorter' than κ ?

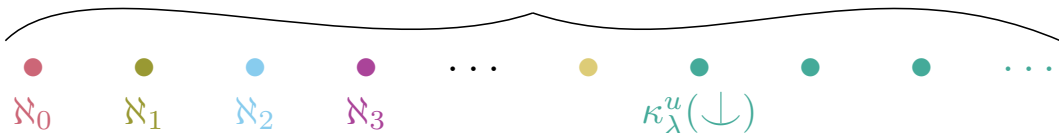
Strictly stable AECs - Main result

Theorem (B., Mazari-Armida)

Suppose $\mathbf{K}$ is an AEC stable in $\lambda \geq \text{LS}(\mathbf{K})$, with amalgamation, joint embedding, and no maximal models, and $\mathbf{K}$ is $\aleph_0$ -tame. Suppose $\perp$ is an independence relation on $\mathbf{K}_\lambda$ that satisfies uniqueness, extension, (universal) continuity, non-forking amalgamation, and $(\geq \kappa)$ -universal local character in some regular $\kappa < \lambda^+$. Let $\kappa_\lambda^u(\perp)$ be the least such κ .

Suppose $N_1, N_2, M \in \mathbf{K}_\lambda$, and $\delta_1 < \delta_2 < \lambda^+$ are regular cardinals, and N_l is a (λ, δ_l) -limit model (over M) for $l = 1, 2$. Then $N_1 \cong N_2$ (fixing M) if and only if $\delta_1 \geq \kappa_\lambda^u(\perp)$.

$\text{REG} \cap \lambda^+$



Applications - First order stable theories

Given first order complete stable theory T , non-forking is a relations satisfying our hypotheses in $\mathbf{K} = (\text{Mod}(T), \preceq)$ with $\kappa_\lambda^u(\perp) = \kappa_r(T) \leq |T|^+$ for all stable $\lambda \geq |L(T)| + \aleph_0$. Thus we have:

Corollary

Let T be a first order complete stable theory. Then for every stable $\lambda \geq |T|$, and any $N_1, N_2, M \in \mathbf{K}_\lambda$ where N_l is a (λ, δ_l) -limit model over M for $l = 1, 2$ where $\delta_1 < \delta_2 < \lambda^+$ are regular cardinals,

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Though the $\Leftarrow$ implication was known, the $\Rightarrow$ implication was largely unexplored.

Strictly stable AECs - Global version

If we assume we have a relation satisfying our conditions at all λ , the cardinal $\kappa_\lambda^u(\perp)$ stabilises for high enough stable λ to a value $\chi(\perp, \mathbf{K}, \leq_{\mathbf{K}}^u)$. This gives the following:

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Theorem (B., Mazari-Armida)

Assume $\perp$ is defined on $\mathbf{K}$ and has all the properties from before, with $\kappa \leq \text{LS}(\mathbf{K})$. Let $\lambda \geq \beth_{(2^{\text{LS}(\mathbf{K})})^+}$ such that $\mathbf{K}$ is stable in λ .

Suppose $\delta_1 < \delta_2 < \lambda^+$ are regular. Then for any $N_1, N_2, M \in \mathbf{K}_\lambda$ where N_l is a (λ, δ_l) -limit model over M for $l = 1, 2$,

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Applications - Algebraic AECs

There are several examples of algebraic AECs satisfying the assumptions of our theorem, but let's focus on one.

Let $\mathbf{K}^{R\text{-mod}}$ be the AEC of modules over a fixed ring R with $\subseteq$. In this case, $\chi(\perp, \mathbf{K}^{R\text{-mod}}, \leq_{\mathbf{K}}^u) = \gamma_r(R)$. In fact, considering when these are $\aleph_0$, we have

Corollary (Originally [Maz21])

The following are equivalent:

1. *R is Noetherian*
2. *for any $\lambda \geq \beth_{(2^{|R|+\aleph_0})^+}$, all λ -limit models in $\mathbf{K}^{R\text{-mod}}$ are isomorphic*

Weaker independence - Existing results

Recall from a few slides back the following ‘long limits are isomorphic’ result outside our setting:

Theorem (Boney, VanDieren)

Let $\mathbf{K}$ be an AEC stable in $\lambda \geq \text{LS}(\mathbf{K})$ with AP, JEP, and NMM in $\mathbf{K}_\lambda$, where $\kappa_\lambda^u(\downarrow_{\lambda\text{-split}}) < \lambda^+$, and $\downarrow_{\lambda\text{-split}}$ satisfies universal continuity and (λ, θ) -symmetry for some regular $\theta \in [\kappa_\lambda^u(\downarrow_{\lambda\text{-split}}), \lambda^+)$.

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Weaker independence - the assumptions

λ -non-splitting is only known to satisfy weakened forms of uniqueness and extension, and the assumed form of symmetry is weakened also. Intuitively, 'universal wiggle room' is needed between the model of the type, and the model it is non-forking over. So our long limit result does not cover this case.

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For example, $\perp$ has *weak uniqueness* if whenever $M_0 \leq_{\mathbf{K}}^u M \leq_{\mathbf{K}} N$ and $p \in \mathbf{gS}(M)$, if $q_1, q_2 \in \mathbf{gS}(N)$, extend p and $\perp$ -do not fork over M_0 , then $q_1 = q_2$.

Weak Indep. - Most general 'long limits are isomorphic' result with AP

The following is (to my knowledge) the most general 'long limits are isomorphic' result assuming AP:

Theorem (B.)

Let $\mathbf{K}$ be an AEC stable in λ , where $\mathbf{K}_\lambda$ has AP. Let $\perp$ be an independence relation on $\mathbf{K}_{(\lambda, \geq \kappa)}$ satisfying weak uniqueness, weak extension, universal continuity in $\mathbf{K}_\lambda$, $(\geq \kappa)$ -local character, and (λ, θ) -weak non-forking amalgamation in some regular $\theta \in [\kappa, \lambda^+)$.*

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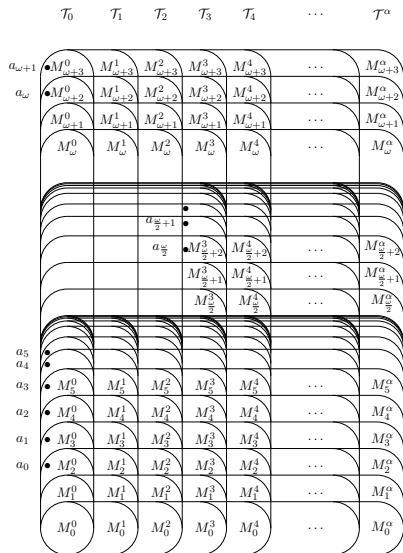
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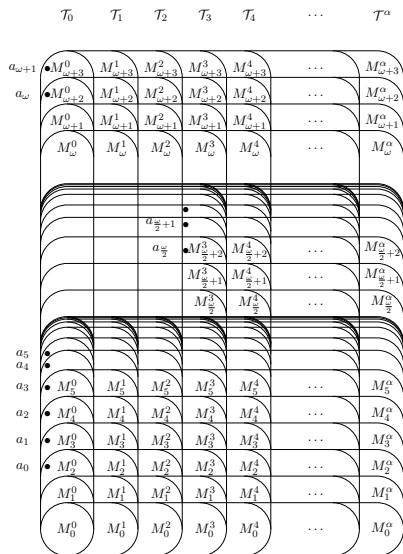
Here universal continuity* in $\mathbf{K}_\lambda$ and (λ, θ) -weak non-forking amalgamations are appropriate weakenings of universal continuity and (λ, θ) -symmetry, so this generalises VanDieren and Boney's result.

Weaker independence - Problems and solutions



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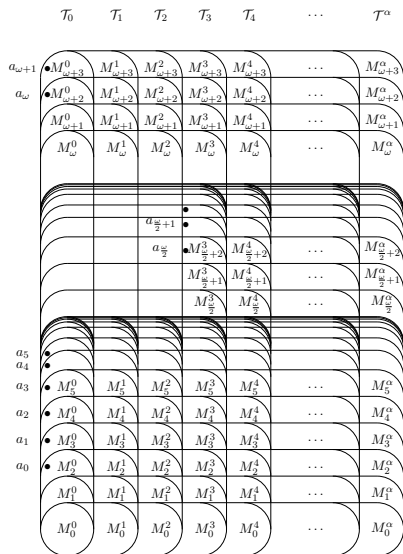
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1. To apply the weak forms of uniqueness, extension, and transitivity, we need 'universal wiggle room'

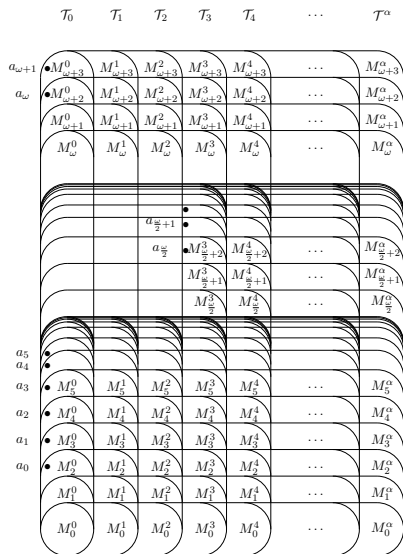
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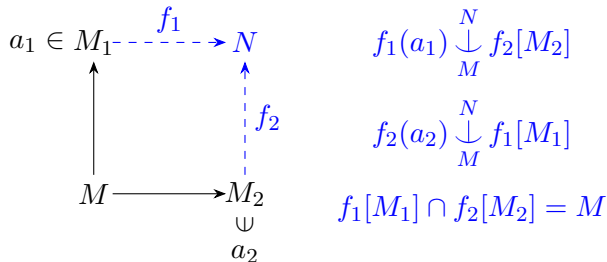
1. To apply the weak forms of uniqueness, extension, and transitivity, we need 'universal wiggle room'
2. In particular, the tower ordering may not be transitive!

To account for these, we modify the definition of tower, and instead of considering single towers, consider *brilliant chains of towers*, and *reduced chains*. With a lot of work, the proof goes through.

DAP - Disjoint non-forking amalgamation

Definition

We say that $\perp$ on $\mathbf{K}$ has *disjoint non-forking amalgamation* if for every $M_1, M_2 \in \mathbf{K}_{\parallel M \parallel}$ such that $M \leq_{\mathbf{K}} M_l$ and $a_l \in M_l$ for $l = 1, 2$, there exist $N \in \mathbf{K}_{\parallel M \parallel}$ and $f_l : M_l \rightarrow N$ fixing M such that $\text{gtp}(f_l(a_l)/f_{3-l}[M_{3-l}], M_3) \perp$ -does not fork over M , and $f_1[M_1] \cap f_2[M_2] = M$.



Theorem (B.)

Let $\mathbf{K}$ be an AEC stable in λ , where $\mathbf{K}_\lambda$ has AP. Let $\perp$ be an independence relation on $\mathbf{K}_{(\lambda, \geq \kappa)}$ satisfying uniqueness, extension, non-forking amalgamation, universal continuity in $\mathbf{K}_\lambda$, and $(\geq \kappa)$ -local character. Then $\perp$ satisfies disjoint non-forking amalgamation.*

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In particular, every $M \in \mathbf{K}_{(\lambda, \geq \kappa)}$ is a disjoint amalgamation base in $\mathbf{K}_\lambda$.

In fact, every $M \in \mathbf{K}_{(\lambda, \geq \kappa)}$ is a disjoint amalgamation base if we only assume weak uniqueness, weak extension, and (λ, θ) -weak non-forking amalgamation for some regular $\theta \in [\kappa, \lambda^+)$.

DAP - The construction

$$\begin{array}{ccc}
 \mathcal{T}^1 & \triangleleft^{(\lambda, \geq \kappa)} & \mathcal{T}^* \\
 M_1 = M_{\alpha,0} & \longrightarrow & M_{\alpha,1} \\
 \vdots & & \vdots \\
 M_{2,0} & \longrightarrow & M_{2,1} \\
 \uparrow & & \uparrow \\
 a_1 \in M_{1,0} & \longrightarrow & M_{1,1} \\
 \uparrow & & \uparrow \\
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 \parallel & & \parallel \wr f \\
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If we sacrifice the ‘non-forking’ requirement, the proof is easier and works with the ‘weak’ assumptions.

Thanks for listening!